

Practical week-3

TASK-1

Develop a C program using fork() that creates a parent and child process. Display the Process ID(PID),Parent Process ID(PPID),and process states at different stages of execution.

```
mohana@MOHANA: ~/task1.c

GNU nano 8.7.1 task1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main(){
pid_t pid;
printf("Parent process started\n");
printf("Parent PID :%d\n",getpid());
printf("Parent PPID :%d\n",getppid());

pid=fork();

if(pid<0){
perror("fork failed");
return 1;
}
else if(pid==0){
printf("\n--- Child Process ---\n");
printf("Child PID : %d\n", getpid());
printf("Child PPID : %d\n", getppid());
printf("Child State: Running\n");

sleep(5);

printf("Child State: Terminating\n");
exit(0);
}

else
{
// Parent process
[ Read 46 lines ]

^G Help          ^O Write Out     ^F Where Is
^K Cut           ^T Execute       ^X Exit
^R Read File     ^\ Replace       ^U Paste
^J Justify
```

```
mohana@MOHANA: ~/task1.c

else
{
// Parent process
printf("\n--- Parent Process ---\n");
printf("Parent PID : %d\n", getpid());
printf("Parent PPID : %d\n", getppid());
printf("Parent State: Running\n");

printf("Parent State: Waiting for child...\n");
wait(NULL);
printf("Parent State: Child terminated\n");
printf("Parent State: Terminating\n");
}

return 0;
}

^G Help          ^O Write Out     ^F Where Is
^K Cut           ^T Execute       ^X Exit
^R Read File     ^\ Replace       ^U Paste
^J Justify
```

```
mohana@MOHANA: ~

mohana@MOHANA:~$ nano task1.c
mohana@MOHANA:~$ gcc task1.c -o task1
mohana@MOHANA:~$ ./task1
Parent process started
Parent PID :656
Parent PPID :356

--- Parent Process ---
Parent PID : 656
Parent PPID : 356
Parent State: Running
Parent State: Waiting for child...
```

```

--- Child Process ---
Child PID   : 657
Child PPID  : 656
Child State: Running
Child State: Terminating
Parent State: Child terminated
Parent State: Terminating

```

TASK-2

To observe the different process states-Ready, Running, Waiting, and Terminated-using Linux monitoring tools such as ps, top, and /proc.

mohana@MOHANA: ~/process_states.c

```

GNU nano 8.7.1 process_states.c
#include <stdio.h>
#include <unistd.h>
int main(){
printf("Process started!\n");
printf("PID: %d\n",getpid());
printf("Running state...\n");
sleep(2);

printf("Waiting/Sleeping state...\n");
sleep(10);

printf("Process terminating...\n");

return 0;
}

```

[Read 15 lines]

^G Help	^O Write Out	^F Where Is
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^J Justify		

mohana@MOHANA: ~

```

mohana@MOHANA:~$ gcc process_states.c -o process_states
mohana@MOHANA:~$ ./process_states
Process started!
PID: 832
Running state...
Waiting/Sleeping state...
Process terminating...

```

Sol:

mohana@MOHANA: ~

```

mohana@MOHANA:~$ ps -o pid,ppid,state,cmd -p 832
  PID   PPID  S CMD
mohana@MOHANA:~$ top
top - 13:56:56 up 51 min,  1 user,  load average: 0.00, 0.02, 0
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 z
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,
MiB Mem : 3758.6 total, 3113.5 free,  462.1 used,  258.
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used. 3296.

  PID USER   PR  NI  VIRT  RES  SHR S  %CPU  %MEM
    1 mohana  20   0 24164 15472 11632 S   0.0   0.4
    2 mohana  20   0  3180  2200  2072 S   0.0   0.1
    7 mohana  20   0  3180  2048  1940 S   0.0   0.1
   56 mohana  19  -1 50428 16784 15484 S   0.0   0.4
   88 systemd+ 20   0 22416 14460 11968 S   0.0   0.4
  103 mohana  20   0 35312 12228  9188 S   0.0   0.3
  120 mohana  20   0  2888  1948  1820 S   0.0   0.1
  121 mohana  20   0  4472  3132  2884 S   0.0   0.1
  123 message+ 20   0  8820  5344  4676 S   0.0   0.1
  155 mohana  20   0 44096 29764 14624 S   0.0   0.8
  162 mohana  20   0 18424  9440  8204 S   0.0   0.2
  177 syslog  20   0 220548 5932  4648 S   0.0   0.2
  220 _chrony  20   0 23892 11340  7136 S   0.0   0.3
  236 _chrony  20   0 12096  2460  1812 S   0.0   0.1
  267 mohana  20   0  5492  2768  2584 S   0.0   0.1
  269 mohana  20   0 123460 32676 18060 S   0.0   0.8
  353 mohana  20   0  3184  1104  980 S   0.0   0.0
  354 mohana  20   0  3200  1244  1108 S   0.0   0.0
  356 mohana  20   0  6268  5528  3856 S   0.0   0.1
  357 mohana  20   0  8644  5536  4696 S   0.0   0.1
  406 mohana  20   0 22340 13272 10992 S   0.0   0.3
  408 mohana  20   0 22916  4076  2076 S   0.0   0.1
  440 mohana  20   0  6136  5416  3864 S   0.0   0.1
  902 mohana  20   0  8164  5996  3896 R   0.0   0.2

```

```
mohana@MOHANA:~$ ./process_states &
[3] 1261
Process started!
PID: 1261
Running state...
mohana@MOHANA:~$ Process terminating...
Waiting/Sleeping state...
ps -o pid,ppid,state,cmd -p 1261
  PID    PPID S CMD
  1261    356 S ./process_states
[2]-  Done                  ./process_states
```